



The electrical and dielectric properties of the Au/Ti/HfO₂/n-GaAs structures

Abdulkerim Karabulut^a, Abdulmecit Türüt^b, Şükrü Karataş^{c,*}

^a Sinop University, Faculty of Engineering, Department of Electrical and Electronics Engineering, Sinop, Turkey

^b Istanbul Medeniyet University, Faculty of Sciences, Engineering Physics Department, TR-34730 Istanbul, Turkey

^c Kahramanmaraş Sütçü İmam University, Faculty of Sciences and Arts, Department of Physics, 46100 Kahramanmaraş, Turkey

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ABSTRACT

In this work, temperature dependent electrical and dielectric properties of the Au/Ti/HfO₂/n-GaAs structures were investigated using capacitance–voltage (C–V) and conductance–voltage (G–V) measurements in the temperature range of 60–320 K by steps of 20 K at 1 MHz. The dielectric constant (ϵ'), dielectric loss (ϵ''), dielectric loss tangent ($\tan\delta$) and ac electrical conductivities (σ_{ac}) have been calculated as a function of temperature. These values of the ϵ' , ϵ'' , $\tan\delta$ and σ_{ac} have been found to be 2.272, 5.981, 2.631 and $3.32 \times 10^{-6} \text{ } (\Omega^{-1}\text{cm}^{-1})$ at 80 K, respectively, 1.779, 2.315, 1.301 and $1.28 \times 10^{-6} \text{ } (\Omega^{-1}\text{cm}^{-1})$, respectively at 320 K. These decrease of the dielectric parameters (ϵ' , ϵ'' , $\tan\delta$ and σ_{ac}) have been observed at high temperatures. The experimental results show that electrical and dielectric properties are strongly temperature and bias voltage dependent.

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1. Introduction

The metal–semiconductor (MS) or metal–insulator–semiconductor (MIS) structures are the basis of a large number of compound semiconductor electronic devices, including microwave diodes, integrated devices, light emitting detectors, field effect transistors, solar cells and photodetectors [1,2]. The MS structures represent essential and basic building blocks of Gallium arsenide (GaAs) based devices as well. Therefore, metal semiconductor contacts are one of the most widely used rectifying contacts in the electronic field [2–6]. The presence of GaAs structure in an active device requires a thorough understanding of the electrical and dielectric characteristics of the structure. GaAs is a compound of the elements gallium (Ga) and arsenic (As). Because of technological importance of metal/GaAs Schottky structures or metal–insulator–semiconductor (MIS) structures, an exact understanding of the nature of the electrical and dielectric properties of these structures is of significant interest in the electronics industry. As an example, for III–V large scale integrated (LSI) devices, a high speed GaAs metal semiconductor field effect transistor (MESFET) by self-aligned gate (SAG) is one of the critical device selections. The

electrical transport in GaAs SBDs has been of considerable interest because of the widespread application of such structures in microwave field effect transistors (FETs), solar cells, and radio-frequency (RF) detectors. Generally, the performance and reliability of a Schottky structure is drastically influenced by the interface quality between the deposited metal and the semiconductor surface [1,2,7–9]. The analysis of electrical and dielectric properties of the Schottky barrier diodes (SBDs) at room temperatures does not only give full information about their conduction process, but also helps to understand the nature of barrier formation at the metal–semiconductor (MS) interface [4–6,10,11]. The C–G–V characteristics depend on different parameters, such as density of interface states, series resistance and barrier height (BH) formation in MS contact. Thus, generally, the performance and reliability of these semiconductor structures depend on the formation of the insulator layer between MS interface and series resistance of structure. The changes in temperatures have important effects in the determination of electrical device characteristics [12–14].

A dielectric is an electrical insulator that can be formed an electric dipole structure. When a dielectric material is placed in an electrical field, through electric charges do not flow through the material, as in a conductor, but only lightly shift from their average equilibrium positions yielding dielectric polarization. The dielectric behavior of materials under external AC field has been the focus of many researches, in view of its critical scientific and technological

* Corresponding author.

E-mail address: skarat@ksu.edu.tr (Ş. Karataş).

importance [15–17]. The temperature dependence electrical and dielectric properties of Au/Ti/HfO₂/n-GaAs MIS structures might be critical for a variety of applications. Thus, the electrical and dielectric properties in GaAs structures have drawn considerable interest in the past decades in electronic and optoelectronic devices [18–21]. The temperature response of the dielectric constant (ϵ'), dielectric loss (ϵ'') and dielectric tangent ($\tan\delta$) is dominated by a low temperature dispersion.

The aim of this our paper is to investigate temperature dependent electrical and dielectric properties such as the dielectric constant (ϵ'), dielectric loss (ϵ''), loss tangent ($\tan\delta$) and AC conductivity (σ_{ac}) of Au/Ti/HfO₂/n-GaAs structure using C–V and G–V measurements in a wide temperature range from 60 K to 320 K at a frequency of 1.0 MHz. Hafnium oxide (HfO₂) is the inorganic insulating compound. Also known as hafnia, this colorless solid is one of the most common and stable compounds of hafnium. HfO₂ is a high dielectric constant ceramic materials, i.e. high-k dielectric. Also, Hafnium oxide reacts with strong acids such as concentrated sulfuric acid as well as with strong bases. Experimental results show that both electrical and dielectric properties of the Au/Ti/HfO₂/n-GaAs MIS structures have considerable correlation with measurement temperature as well as applied bias voltage.

2. Experimental details

The Au/Ti/HfO₂/n-GaAs structures were fabricated using n-type GaAs wafer with (100) crystal orientation, having thickness of 300 μm , 2 in. diameter, $6.8 \times 10^{15} \text{ cm}^{-3}$ carrier concentration (N_D) and 1.2 $\Omega \text{ cm}$ resistivity. Before the SBD fabrication processes, the experimental procedures were carried out first as cleaning procedure consisting of; sonicated for 2 min in acetone and propanol, subsequently. Then, rinsed in deionized (DI) water thoroughly and after dried under N₂ flow. Thus, after surface cleaning of n-GaAs, high purity (99.999%), In with a thickness of about 2000 Å (Å) was deposited for back-side ohmic contact at a reactor pressure about 10^{-6} Torr in a high vacuum thermal evaporation system. To form the low-resistance ohmic contact, In-deposited n-GaAs wafer was annealed at 385 °C for 3 min under the N₂ atmosphere. After this process, HfO₂ thin films were deposited on cleaned top surface of GaAs samples using standard thermal atomic layer deposition (ALD) system (Savannah S100 ALD reactor, Cambridge Nanotech Inc.). Deposition was performed at 200 °C using tetrakis (dimethylamido hafnium [Hf(NMe₂)₄] and water (H₂O) as the organometallic precursor and oxygen source, respectively. Hf(NMe₂)₄ was preheated to 75 °C and stabilized at this temperature before the deposition. Carrier gas (N₂) with a flow rate of 20 sscm was used. Pulse times of the Hf precursor and oxygen source were 0.2 and 0.015 s, respectively. Purging was performed after each step. Deposition rate for this recipe was determined about 1 Å per cycle. The total number of ALD cycles was used to precisely control the thickness of deposited hafnia layer. Afterwards, for Schottky diode device fabrication, hafnia coated and In-back-contacted GaAs sample was patterned via standard lithography process. After development process, sample was rinsed, entirely, in de-ionized water. On top of the ALD-grown HfO₂ insulating layer, Au (90 nm)/Ti (10 nm) Schottky contacts were formed using dc magnetron sputtering technique. Finally, top metal Schottky contacts were defined using a standard lift-off process. After fabrication process, The C–G–V–T characteristics were measured using a HP model 4192A LF impedance analyzer at room temperature in dark environment.

3. Results and discussion

The temperature dependent electrical and dielectric properties

of Au/Ti/HfO₂/n-GaAs structures are presented and discussed in this section.

3.1. Temperature dependence of electrical properties

Figs. 1 and 2 show the measured C–V and G/ω–V characteristics of the Au/Ti/HfO₂/n-GaAs structure as a function of forward and reverse voltage within the temperature range of 60–320 K, respectively. As shown in Figs. 1 and 2, the capacitance-voltage (C–V) and conductance-voltage (G/ω–V) plots depend on temperature. The C–V characteristics were measured in +1.0 to –3.0 V range at each temperature, where the forward voltage ranges from 0.0 to 0.10 V, and the reverse voltage ranges from 0.0 to –3.0 V. In Fig. 1, the decrease in capacitance in the reverse bias regime from 0.0 to –3.0 V implies an increase in the semiconductor depletion width. Since the charge neutrality conditions at the interface state should be satisfied, widening of the depletion width in the reverse bias C–V characteristics results from a reduction of the ionized donor concentration [22]. The C value increases with decreasing temperature at a given bias voltage (from 0.0 to 0.1 V). This result can be attributed to the fact that the interface states following the AC signal at low frequencies and therefore an excess capacitance occurs in addition to the space charges capacitance because of the contribution of interface charges [23–26]. Furthermore, as shown in Fig. 2, the G/ω–V characteristics of Au/Ti/HfO₂/n-GaAs structure increased with temperature depending on the applied voltage, whereas the values of conductance decreased with increasing temperature above 1.0 V forward bias. An intersection point is observed at about 1.0 V in the temperature-dependent G/ω–V characteristics (Fig. 2). Such a behavior is attributed to reordering of the interface state densities and series resistance. Alternative, this specific G/ω– behavior might be attributed to a particular distribution of the interface states between metal-semiconductor layers [17,27, and 28]. Thus, it can be said that the C_m and G/ω plots are both temperature and voltage dependent. A behavior of capacitance and conductance is primarily attributed to particular distribution of surface states at the HfO₂ layer GaAs semiconductor

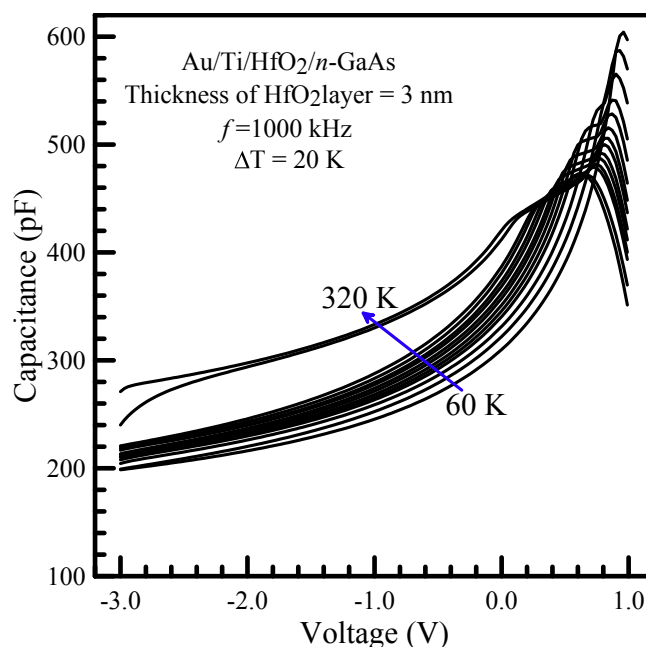


Fig. 1. The C–V curves for the Au/Ti/HfO₂/n-GaAs structure at different temperatures and 1000 kHz.

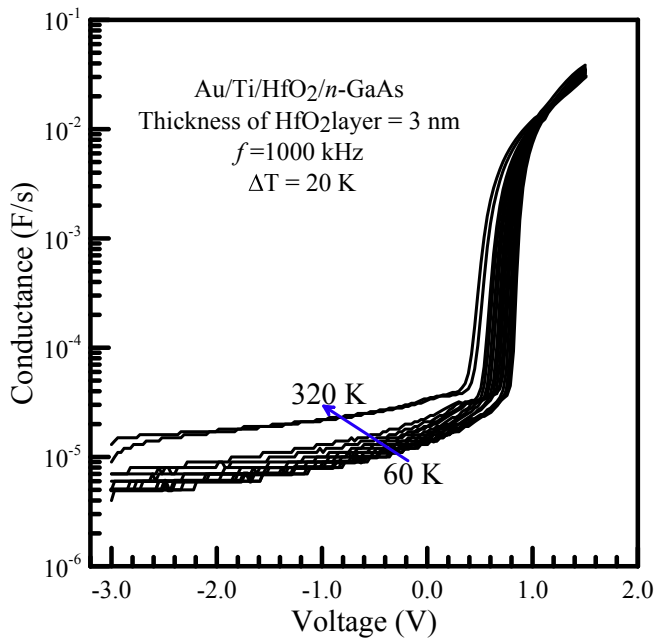


Fig. 2. The G/ω – V curves for the Au/Ti/HfO₂/n-GaAs structure at different temperatures and 1000 kHz.

interface. The variations of C – V and G/ω – V characteristics with temperature at high frequency (1000 kHz) at which only the free carriers within the majority bands were able to respond to the small excitation ac signal are shown in Figs. 1 and 2 for Au/Ti/HfO₂/n-GaAs structures, respectively [17]. In addition, as shown in Figs. 1 and 2, the values of both capacitance and conductance increase with temperature. Fig. 3 shows the reverse and forward bias C^{-2} – V curves from the data in the targeted temperature range. In the fabricated Au/Ti/HfO₂/n-GaAs MIS structure the capacitance of the barrier varies with applied reverse voltage in such a way that a

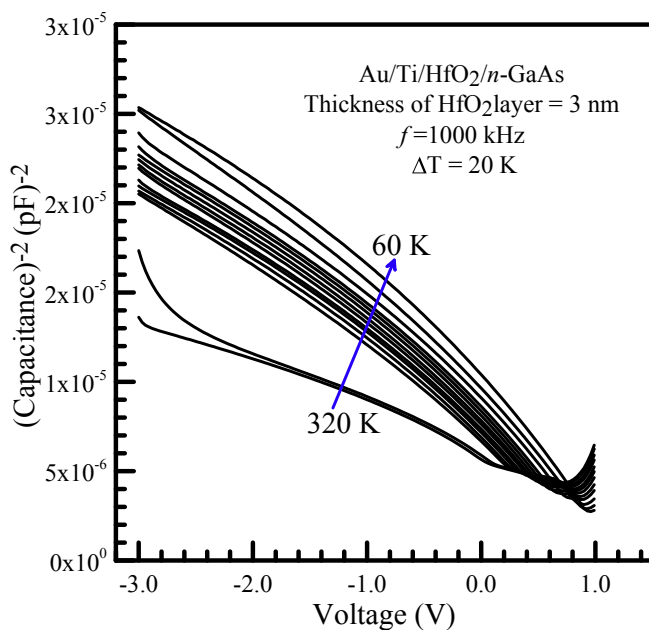


Fig. 3. The reverse bias C^{-2} – V characteristics for the Au/Ti/HfO₂/n-GaAs structure at different temperatures and 1000 kHz.

straight line is obtained if C^{-2} is plotted against voltage (V). As can be seen in Fig. 3, the C^{-2} versus V plots is linear in a certain voltage ranges.

Fig. 4 show measured impedance modulus-voltage (Z – V) characteristics of Au/Ti/HfO₂/n-GaAs structure as function of reverse and forward voltage at 1000 kHz. The impedance modulus characteristics were measured in +0.10 to –3.0 V range for each temperature. As seen from Fig. 4, the impedance modulus plots depend on voltage and temperature and decreases with increasing temperatures depending on the applied voltage, whereas the values of capacitance and conductance increased with temperature. Thus, the impedance transition voltage and the ratios of the impedances drop after transition are very similar [5,29,30].

3.2. Temperature dependence of dielectric constant (ϵ'), dielectric loss (ϵ''), loss tangent ($\tan\delta$) and ac electrical conductivity (σ_{ac})

In metal insulator semiconductor structures, as dependence to applied voltage is distributed between dielectric layers, depletion layer and recharged surface states [31–33]. Therefore, dielectric constant is strong indicator of the property and electrical states of the material. Temperature dependence of dielectric constant (ϵ'), dielectric loss (ϵ''), loss tangent ($\tan\delta$) and electrical conductivity (σ_{ac}) were obtained from the data of capacitances and conductance's measurements for the Au/Ti/HfO₂/n-GaAs structure in the same temperature range of 60–320 K. Basic dielectric parameters such as dielectric constant (ϵ'), dielectric loss (ϵ''), loss tangent ($\tan\delta$) and ac electrical conductivity (σ_{ac}) are expressed as in the following equations [17,34–36],

$$\epsilon^*(\omega) = \epsilon'(\omega) - i\epsilon''(\omega) \quad (1)$$

where ϵ' and ϵ'' are real and imaginary of complex permittivity (ϵ^*), $\omega = 2\pi f$, and f is the frequency of the applied electrical field. The part with real of the dielectric constant, ϵ' , is a measure of the energy charged from the applied electrical field in the material and identifies the strength of alignment of dipole in the dielectric. The

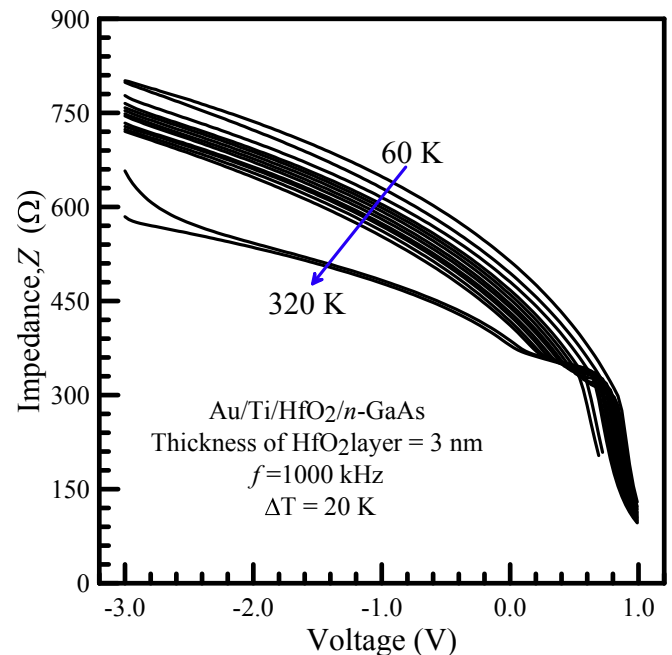


Fig. 4. Experimental forward and reverse bias impedance modulus-voltage characteristics for the Au/Ti/HfO₂/n-GaAs structure at different temperatures and 1000 kHz.

part with imaginary, ϵ'' , or loss factor, is the energy scattered in the dielectric associated with the frictional dampening that prevent displacements of bound charge from remaining in phase under electrical field variation [37]. The complex permittivity formalism have been employed to describe the electrically and dielectric properties. In the ϵ' formula, in the case of admittance (Y^*) measurements, the following relation holds [36,37];

$$\epsilon^* = \frac{Y^*}{j\omega C_0} = \frac{C_m}{C_0} - j \frac{G}{\omega C_0} \quad (2)$$

where C_m and G are the measured capacitance and conductance of the structure. The part with real of the dielectric constant (ϵ') at different temperatures is calculated using the measured capacitance values (C_m) at the strong accumulation region according to the relation [38–40];

$$\epsilon' = \frac{C_m}{C_0} \quad (3)$$

where $C_0 = (\epsilon_0 A / d_{ox})$; A is the area of the sample, d_{ox} is the interfacial layer thickness and ϵ_0 is the permittivity of free space charge, and C_m is measurement maximal capacitance of structure in the strong accumulation region, correspond to the oxide capacitance. The imaginary part, or loss factor (ϵ'') is expressed as,

$$\epsilon'' = \frac{G_m d_{ox}}{A \epsilon_0 \omega} \quad (4)$$

where G_m is the conductivity of device, and also, the part with imaginary is expressed as,

$$\epsilon'' = \epsilon' \tan \delta \quad (5)$$

and thus loss tangent,

$$\tan \delta = \frac{\epsilon''}{\epsilon'} \quad (6)$$

Furthermore, temperatures dependent ac electrical conductivity, $\sigma_{ac}(\omega)$, was obtained directly from the loss tangent factor as;

$$\sigma_{ac}(\omega) = 2\pi f \epsilon_0 \epsilon' \tan \delta = \omega \epsilon_0 \epsilon' \tan \delta \quad (7)$$

The electrical conductivity provides a measure of the mobility of charge when an electrical potential is placed across a structure. Figs. 5–7 show the variation of the dielectric constant (ϵ'), dielectric loss (ϵ''), loss tangent ($\tan \delta$) and AC electrical conductivity (σ_{ac}) at high frequency (1 MHz) at different temperatures for the Au/Ti/HfO₂/n-GaAs structure, respectively. From the measured values of capacitance and conductance, the dielectric properties were found to be strongly dependent to temperature. The temperature dependent of ϵ' , ϵ'' , $\tan \delta$ and σ_{ac} profiles were obtained using Eqs. (3)–(7), and are given in Figs. 5–7, respectively. The observed change in dielectric properties can be understood by considering the displacement damage introduced by temperatures in the decrease in the values of ϵ' , ϵ'' and $\tan \delta$. It can be seen in Figs. 6–8, the values of ϵ' , ϵ'' and $\tan \delta$ decrease with increasing temperatures. As mentioned in ref. 39–42, the decrease in ϵ' , ϵ'' and $\tan \delta$ with increase in temperature is explained by the fact that as the temperature raised, the interfacial dipoles have less time to orient themselves in the direction of the alternating field. Also, the surface states in equilibrium with the semiconductor can easily follow the AC signal at low temperatures and an excess ϵ' , ϵ'' and $\tan \delta$ occur in addition to real dielectric ϵ' , ϵ'' and $\tan \delta$ [43,44]. Similar studies have been also reported in literature as well [17,31–44].

The AC electrical conductivity (σ_{ac}) curve of diode against

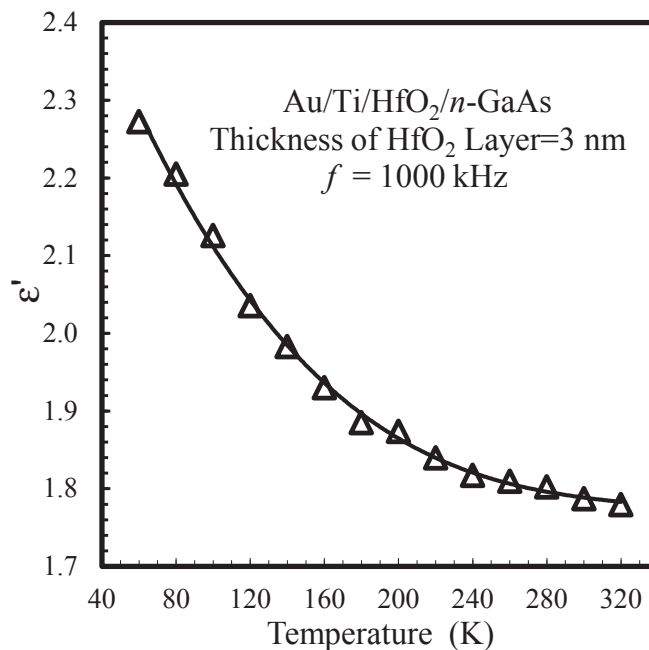


Fig. 5. Temperature dependence of dielectric constant (ϵ') of the Au/Ti/HfO₂/n-GaAs structure at 1000 kHz.

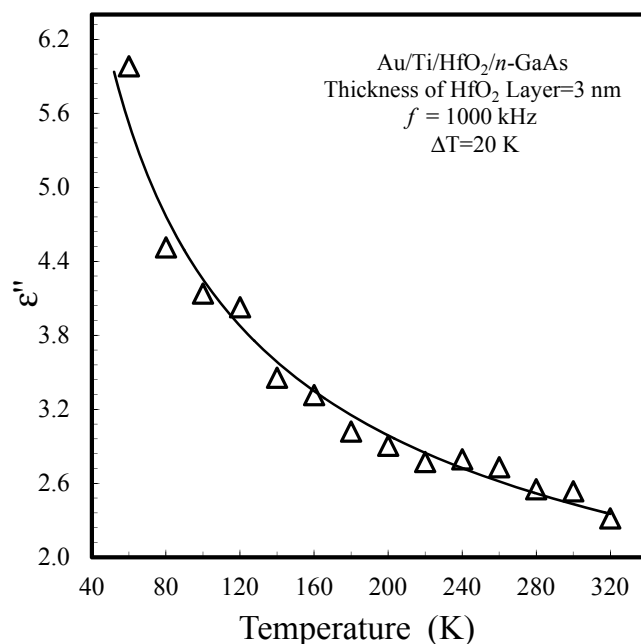


Fig. 6. Temperature dependence of dielectric loss (ϵ'') of the Au/Ti/HfO₂/n-GaAs structure at 1000 kHz.

temperatures in the range from 60 K to 320 K temperature at a constant frequency (1000 kHz) is presented in Fig. 8. As shown from this figure; as other dielectric properties, the electrical conductivity decreases with increasing temperature. It is observed that the electrical conductivities decreases as the temperature increase from 60 K to 320 K. The decrease in electrically conductivity with the increasing temperature may be because of the increase in the thermally activated drift mobility of charge carriers according to the hopping conduction mechanism. In such a hopping process the carrier mobility is temperature dependent, characterized by an

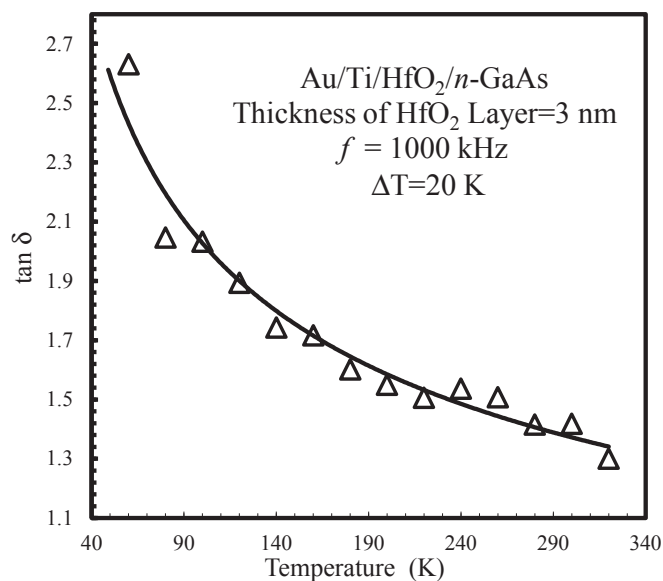


Fig. 7. Temperature dependence of loss tangent ($\tan\delta$) of the Au/Ti/HfO₂/n-GaAs structure at 1000 kHz.

activation energy, which corresponds to the energy picture of the crystal lattices around the area of electrons [45,46]. Thus, this behavior can be attributed to a gradual decrease in series resistance with increasing temperature [17]. On the other hand, the decrease of the electrical conductivity leads to a drop in the eddy current which in turn decreases the energy loss. It is clearly seen that the process of dielectric polarization in the HfO₂ layer- GaAs semiconductor structure takes place through a mechanism similar to the conduction process.

Fig. 9 shows the $\ln\sigma$ versus $1000/T$ plot of the diode. As shown in Fig. 9 a plot of $\ln\sigma$ versus $1000/T$ is a straight line. The linear relationship between the total conductivity and absolute temperature can be given by the following equation [46,47];

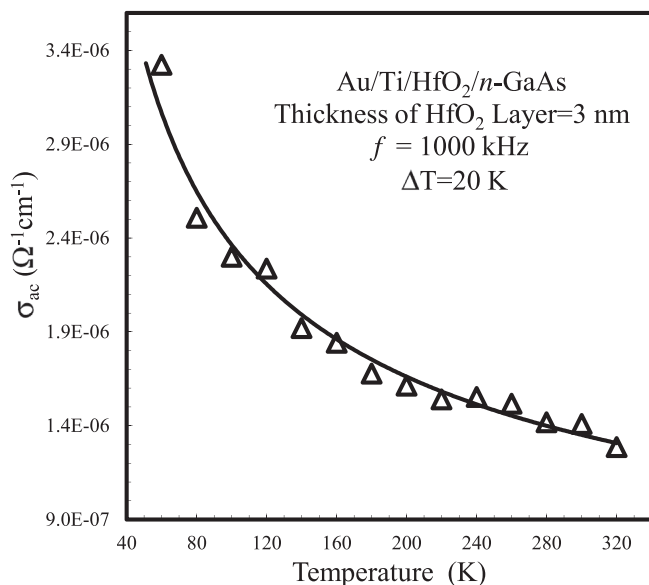


Fig. 8. Temperature dependence of ac electrical conductivity (σ_{ac}) of the Au/Ti/HfO₂/n-GaAs structure at 1000 kHz.

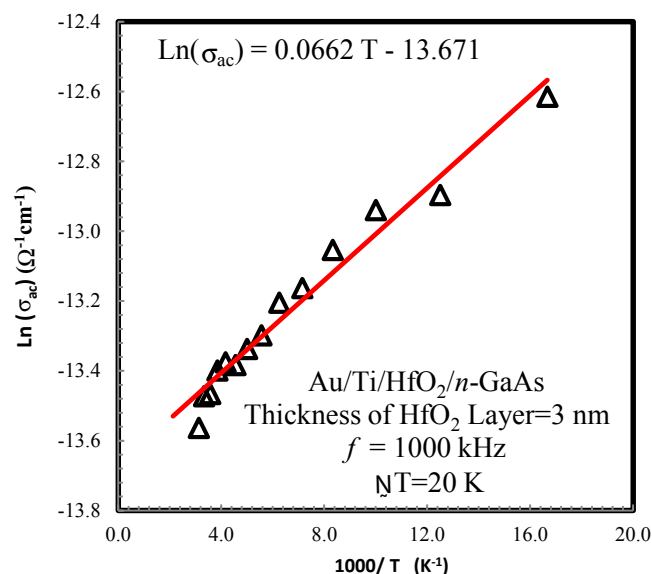


Fig. 9. ac electrical conductivity (σ_{ac}) versus $1000/T$ of the Au/Ti/HfO₂/n-GaAs structure at 1000 kHz.

$$\sigma_{ac} = \sigma_o \exp\left(\frac{-E_a}{kT}\right) \quad (8)$$

where σ_o represents the composite constant and E_a is the activation energy. Also, Fig. 9 is an Arrhenius curve, and the activation energy is found to be 0.0662 eV from the slopes of σ_{ac} versus $1000/T$ plot. The small activation energy shown that the bulk trap level lies above the intrinsic fermi level and is very close to the conduction band. Furthermore, such low activation energy at this temperatures can be attributed to the existence of charges at traps [46–48]. Thus, value of activation energy obtained above is not in consistence with the reported values for Au/n-type GaAs Schottky diodes [1,2, and 9]]. This means that these small activation energies shown that the bulk trap level lays above the intrinsic fermi levels and is very close to the conduction band [48]. As conclusion, low activation energies can be attributed to the existence of charges at traps.

4. Conclusions

In this study, general electrically and dielectric properties of the MIS device have been reported in detail from C–V and G/ω –V measurements in the temperature range of 60–320 K by steps of 20 K at 1000 kHz. It has been shown that the values of capacitance-voltage and conductance-voltage increase with temperature while impedance-voltage show reverse temperature dependence. The experimental studies shown that both the capacitance-conductance-voltage and impedance-voltage characteristics have been quite sensitive to temperature and applied bias voltage. These behaviors are attributed to the interface surface states which can easily follow the AC signal at low temperatures. In addition, we have investigated temperature-dependent dielectric properties of the Au/Ti/HfO₂/n-GaAs structure, and we can conclude that the dielectric values (ϵ' , ϵ'' , $\tan\delta$ and σ_{ac}) of Au/n-GaAs structure strongly depends on temperature. The decreases in dielectric values with increasing temperatures can be attributed to charge centers created. That is, the observed change in electrical and dielectric properties can be understood by according to displacement damage introduced by temperature and applied voltage. The observations observed in this study might be used for MIS contacts in integrated circuits and other electronic devices.

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